

Summaiya Bibi

Software Engineer

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PROFESSIONAL SUMMARY

Results-driven and detail-oriented Software Engineer with 3+ years of experience in Programming Languages & Databases: Python, Machine Learning & AI: scikit-learn, TensorFlow, PyTorch, Natural Language Processing (NLP), Computer. AI/ML Engineer with hands-on experience in developing Machine Learning, Deep Learning, and Generative AI solutions using Python and SQL. Skilled in predictive modeling, Natural Language Processing (NLP), Computer Vision, Retrieval-Augmented Generation (RAG), and AI agents. Experienced in data preprocessing, feature engineering, model evaluation, REST API development, and deploying intelligent applications using FastAPI, Flask, and Streamlit. Passionate about building scalable AI solutions and continuously learning emerging AI technologies. Recognized for delivering scalable solutions, optimizing system performance, and driving measurable business impact.

TECHNICAL SKILLS

Programming & Core: Programming Languages & Databases: Python, Computer, Docker, SQL, MySQL, Frameworks & Tools: FastAPI, Streamlit

AI / Machine Learning & Frameworks: Machine Learning & AI: scikit-learn, TensorFlow, PyTorch, Natural Language Processing (NLP), Generative AI: LangChain, Retrieval-Augmented Generation (RAG), OpenCV

Cloud, Databases & DevOps: Flask, Git

Analytics, Tools & Methodologies: Vision, AI Agents, Pandas, NumPy

PROJECTS

- Heart Disease Risk Detection System (Final Year Project) | Python, Flask, XGBoost, Scikit-Learn, HTML/CSS, Pandas Designed and Developed an end-to-end medical risk prediction pipeline, benchmarking 4 Machine Learning models (XGBoost, Random Forest, SVM, Logistic Regression) to select the optimal classifier achieving 90%+ accuracy. Built and deployed a production-ready Flask REST API to handle data preprocessing, feature engineering, cross-validation, and real-time model inference. Designed an intuitive web application interface allowing non-technical users and healthcare personnel to execute dynamic cardiovascular risk assessments.
- Live AI Assistant & Task Engine | Python, LangChain, OpenAI API, Speech Recognition Developed a voice-enabled AI agent integrating LLM APIs and speech-to-text modules for real-time conversational intelligence and automated workflow execution.
- Enterprise Document Query Engine (GenAI / RAG) | Python, LangChain, FAISS Vector DB, FastAPI Designed a Retrieval-Augmented Generation (RAG) system using LangChain and FAISS for semantic question answering over unstructured PDF documents through FastAPI.
- Computer Vision Object Detection Pipeline | Python, OpenCV, PyTorch, YOLO Implemented a real-time object tracking and detection deep learning workflow using YOLO and PyTorch, incorporating optimized frame preprocessing for low-latency video analytics.
- Deployed Movie Recommendation Engine | Python, NLP, TF-IDF, Cosine Similarity, Streamlit Cloud Engineered and deployed a content-based recommendation system utilizing TF-IDF vectorization and cosine similarity for personalized, real-time movie suggestions.

EDUCATION

- BS Computer Science — Virtual University of Pakistan (2021 – 2025)

CERTIFICATIONS

- Deep Learning Specialization (DeepLearning.AI)
- AWS Certified Machine Learning Specialty